

# Running the CalGames Content Desk

A complete setup and operating guide for volunteers

**CalGames 2026 · Woodside High School · October 16-18 · Western Region Robotics Forum**

# Read this first

This guide assumes you have never run this program, never used OBS, and never touched a video switcher. That is the expected starting point. Nothing here requires you to type commands, edit code, or understand how any of it works.

**You do not need to read all of it.** Find your job below and read that chapter plus the two short ones before it. The whole thing takes about forty minutes end to end; your chapter takes ten.

| IF YOU ARE...                                           | READ CHAPTERS  |
|---------------------------------------------------------|----------------|
| Setting up on Friday                                    | 1, 2, 3, 4, 5  |
| <b>Desk manager / producer</b>                          | 1, 2, 6, 7, 11 |
| <b>On-air talent</b> (announcing, analysis, interviews) | 1, 2, 8        |
| <b>Field tech</b>                                       | 1, 2, 3, 9     |
| Replay operator                                         | 1, 2, 7        |
| Trivia host or arcade scorer                            | 1, 2, 10       |
| Anyone who just needs a screen on a TV                  | 1, 2, 5        |

**The one rule.** If something looks wrong on the broadcast, the fix is almost never in this program. It is a person pressing the wrong button, and the person who notices should say so out loud. Nobody gets in trouble for calling out a wrong graphic. The show is a team sport.

## Chapter 1 · What this thing is

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The content desk is one program that runs on one laptop. Everything else (every screen, every control panel, every phone) is just a web page pointed at that laptop.

That is the single most useful idea in this guide, so it is worth saying twice:

**There is one program. Everything else is a browser tab.**

So there is nothing to install on the pit monitors, nothing to install on the announcer's tablet, and nothing to install on anyone's phone. You type an address, and a screen appears.

### What it does

- Draws the **scoreboard and graphics** that go on the stream and on the TVs in the hall.
- Tells the **announcer** who is playing, what their record is, and what they need to do to earn a ranking point. In words, not jargon.
- Cuts **replays** and **match videos**, and uploads them.
- Runs the **crowd trivia** and the **arcade** side tournament between matches.
- Runs the **house music** in the hall.
- Answers "**when does my team play?**" on any parent's phone.

### The three jobs

At minimum this show runs with three people. More is better; three is enough.

**Desk manager (producer).** Sits at the desk laptop. Drives the graphics, switches cameras, keeps the show moving. If there is only one person who knows how the desk works, it is this one.

**On-air talent.** Announces, analyses, does interviews. Works off a tablet showing the talent view, which tells them everything they need to say.

**Field tech.** Owns the connections: is the desk talking to the field, is OBS up, is the stream live, is the recording running. Checks the pre-show list in chapter 3 and fixes what it flags.

If you have more people, the next two to add are a **dedicated switcher** (so the desk manager stops cutting cameras) and a **second talent** (so the announcer has somebody to talk to). After that, a replay operator.

## Chapter 2 · Starting the desk

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### Step 1: Copy one file

Copy `Ca1GamesContentDesk.exe` onto the desk laptop. Put it anywhere: the Desktop is fine.

That is the whole installation. There is no installer, no admin password, and nothing goes into Program Files.

### Step 2: Double-click it

A black window opens and prints what it is doing. It takes about a minute the first time and a few seconds after that.

Windows may show a blue "**Windows protected your PC**" box. This is normal for any program that has not paid for a code-signing certificate. Click **More info**, then **Run anyway**. If you are not comfortable doing that, get the content desk lead. Do not click past a security warning on somebody's word alone, including this document's.

### Step 3: Read the address

When it is ready the window prints something like:

```
READY
http://10.0.100.23:8720/
PIN 0864
```

**Write that address on the whiteboard.** Every screen in the building is going to need it. Write the PIN there too. It is a door code, not a secret; roughly eight people need it.

The numbers will be different at your event. Use what your window prints, not what is printed here.

### Step 4: Leave the window open

Closing the window stops the desk and every screen in the building goes blank. Minimise it, push it to a second monitor, but do not close it.

To stop the desk deliberately, close the window or press **Ctrl-C** in it. Either one shuts down cleanly and releases the address so the next start works.

## If it will not start

| WHAT YOU SEE                           | WHAT TO DO                                                               |
|----------------------------------------|--------------------------------------------------------------------------|
| "Port already in use"                  | An older copy is still running. The launcher offers to stop it. Say yes. |
| It prints a field-scan failure         | Fine. The desk runs without the field; see chapter 9.                    |
| Nothing happens at all                 | The file did not finish copying. Copy it again.                          |
| A blue "Windows protected your PC" box | More info → Run anyway. See step 2.                                      |

## Trying it at home first

You do not need a field, a camera, or an event to practice. When the desk finds no field it asks whether you want a **pretend match**; press **D** and every screen plays a full match on a loop with realistic scores.

After the first run, two files appear in the desk folder:

- **START-DESK.cmd**: the real thing.
- **START-PRACTICE.cmd**: the pretend match.

Both are double-click. Practice on the second one as much as you like; it cannot touch a real field or upload anything.

# Chapter 3 · The pre-show check (field tech)

Open the desk address in a browser on the desk laptop. You get an index page listing every screen. Open **Desk console** ( /s/desk ). It asks for the PIN once.

At the top of the desk console is the **doors check**. It answers one question: *is this thing ready?* Everything on it is green, amber, or red.

Run it **at doors**, meaning when the audience is let in, not five minutes before the first match.

| CHECK       | WHAT RED MEANS                                                   | WHAT TO DO                                            |
|-------------|------------------------------------------------------------------|-------------------------------------------------------|
| Field       | The desk cannot see the scoring system                           | Chapter 9                                             |
| OBS         | The desk cannot drive the camera switching                       | Chapter 4                                             |
| Recording   | Nothing is being recorded, so there will be no replays or videos | Check the capture device is plugged in                |
| Disk        | Less than two hours of recording space                           | Free up space or change disks now, not at lunch       |
| Publishing  | YouTube is not set up                                            | Videos will not upload. Talk to the content desk lead |
| Microphones | A mic is muted in OBS                                            | Unmute it                                             |
| House audio | The music machine is not connected                               | Chapter 10                                            |

**Amber is not red.** Amber means "this will work, but you should know". A missing Blue Alliance key is amber: videos still upload, they just do not get linked.

Everything red on this list will be noticed by somebody in the audience within an hour. That is the entire point of running it early.

## Chapter 4 · Setting up OBS

OBS Studio is the free program that actually sends the video to YouTube. The content desk does not stream anything itself: it draws the graphics, and OBS puts them on top of the camera picture.

If somebody else has already set up OBS at this event, skip to chapter 5.

### Step 1: Install OBS

Download OBS Studio from **obsproject.com** and install it. Take every default.

### Step 2: Make the scenes

A "scene" in OBS is one arrangement of pictures: a camera, a graphic, a title. You switch between scenes to change what the audience sees.

Create six scenes with **exactly these names**. Capitalisation and underscores matter, because the desk switches them by name:

| SCENE NAME | WHAT IT SHOWS                                |
|------------|----------------------------------------------|
| CG_INTRO   | The opening title, before the day starts     |
| CG_MATCH   | The field camera, the main one               |
| CG_SCORE   | The field camera, for the final score reveal |
| CG_REPLAY  | The replay playback                          |
| CG_DESK    | The analysis desk camera                     |
| CG_ARCADE  | The video game capture, for between matches  |

To add a scene: in the **Scenes** box at the bottom left, click +, type the name, press Enter.

If you get a name wrong, the desk's camera buttons will do nothing and nothing will explain why. Check the spelling twice.

### Step 3: Add the match overlay

This is the important one. It is the scoreboard.

1. Select the **CG\_MATCH** scene.
2. In the **Sources** box, click + → **Browser**.
3. Name it `over1ay` and click OK.
4. Fill in the box exactly like this:

| FIELD                                     | VALUE                                                                    |
|-------------------------------------------|--------------------------------------------------------------------------|
| URL                                       | <code>http://10.0.100.23:8720/s/program</code> (use <i>your</i> address) |
| Width                                     | <code>1920</code>                                                        |
| Height                                    | <code>1080</code>                                                        |
| Use custom frame rate                     | leave unticked                                                           |
| Shutdown source when not visible          | <b>UNTICK THIS</b>                                                       |
| Refresh browser when scene becomes active | <b>UNTICK THIS</b>                                                       |

Those last two matter. If either is ticked, the overlay reloads every time you cut back to the camera, and the audience watches the scoreboard flash and re-animate mid-match.

1. Click OK. The scoreboard appears over your camera.
2. Drag it so it fills the frame exactly. Right-click it → **Transform** → **Fit to screen** does this in one step.

#### Step 4: Copy the overlay to the other scenes

The scoreboard belongs on the score scene and the replay scene too.

Do **not** add a second Browser Source: that runs a second copy of the overlay and doubles the load on the laptop. Instead:

1. Right-click the `Overlay` source → **Copy**.
2. Select `CG_SCORE`, right-click in the Sources box → **Paste (Reference)**.
3. Do the same for `CG_REPLAY`.

**Paste (Reference)**, not **Paste (Duplicate)**. A reference is the same overlay showing in two places. A duplicate is two overlays, and they will drift out of step with each other.

#### Step 5: The telestrator (optional)

If your analyst is going to draw on replays:

1. Select `CG_REPLAY`.
2. + → **Browser**, name it `Telestrator`.
3. URL `http://YOUR-ADDRESS/s/tele`, 1920 × 1080, same two boxes unticked.
4. Drag it **above** the Overlay in the Sources list.

#### Step 6: The arcade overlay (optional)

Same recipe on the `CG_ARCADE` scene, with URL `/s/arcade`.



## Step 7: Let the desk drive OBS

This lets the desk manager cut cameras from the desk console instead of reaching for OBS.

1. In OBS: **Tools** → **WebSocket Server Settings**.
2. Tick **Enable WebSocket server**.
3. Click **Show Connect Info** and note the **Server Password**.
4. Click OK.

Then tell the desk about it. Ask the content desk lead to start the desk with OBS enabled. This is one of the few things that needs a person who is comfortable with a command line, and it is a two-minute job.

You can run the entire show without this. The desk manager just switches scenes in OBS by hand instead. Everything else works identically.

## Step 8: Check it

Load a practice match (chapter 6) and look at your OBS preview. You should see the scoreboard over the camera, with the clock counting.

If the overlay is **black instead of transparent**, the Browser Source lost its transparency. Delete it and add it again. Do not add a Color Correction filter, which is the usual wrong fix.

## A note for events with a hardware switcher

If you are going through an ATEM or similar rather than OBS, add `?key=luma` to the overlay URL:

```
http://YOUR-ADDRESS/s/program?key=luma
```

That swaps the transparent background for solid black so a luma key works. Without it the graphics will key out into nothing. On the OBS path, leave it off.

## Chapter 5 · The screens around the hall

Every TV, monitor and phone in the building shows a web page. None of them need the PIN.

### The easy way

On the monitor, open a browser and go to the desk address. Click **Pick a screen**. Choose from the list. Press **F11** for full screen.

That is it. The page remembers full screen, so if the monitor loses power and comes back, F11 once and it is right again.

### The screens, and where each one belongs

| SCREEN                      | ADDRESS                 | PUT IT                                                                                           |
|-----------------------------|-------------------------|--------------------------------------------------------------------------------------------------|
| Program with the field feed | /s/watch?screen=program | Pit monitors, concession TVs: the broadcast picture without needing OBS                          |
| Side screen                 | /s/watch?screen=side    | The big TVs beside the field. Shows who is on deck and the current rankings, rotating on its own |
| When do we play?            | /s/watch?screen=next    | Anywhere parents gather. Also the QR code on the side screens                                    |
| Trivia                      | /s/watch?screen=trivia  | A monitor the crowd can see between matches                                                      |
| Arcade                      | /s/watch?screen=arcade  | Over the video game capture                                                                      |

### Side screens specifically

The side screens are the ones most people in the hall will actually look at. They rotate on their own between "who plays next" and the rankings, with no operator.

They are designed to be read from thirty feet away, which means the type is large and there is not much on screen at once. That is deliberate. Do not try to fit more on.

If a side screen goes blank or shows "reconnecting", the desk laptop went away. The screen recovers by itself when the desk comes back; you do not need to touch the monitor.

### Checking a display with the test card

Before trusting any TV, projector, or LED wall, open ADDRESS/s/testcard on it, full screen. Four looks and you are done:

- **All four corner marks visible?** If not, the display is cropping the edges, and it will crop the scoreboard the same way.

- **Is the circle round?** An ellipse means the picture is stretched.
- **Can you tell the dark gray steps apart?** If the left end of the gray ramp merges into black, the display will crush the broadcast's purple too.
- **Is the clock ticking?** A frozen clock means a frozen signal.

The dashed rectangles are the exact safe areas the graphics keep to, so whatever the display does to them, it does to the score bar.

## The field feed behind the program screen

`/s/watch?screen=program` shows the overlay on top of a live picture of the field. That picture comes from a setting the content desk lead fills in ( `kiosk.fieldStreamUrl` ). If it is blank you get the overlay on the CalGames backdrop instead, which looks intentional and is fine.

# Chapter 6 · The desk console (desk manager)

Open `/s/desk` . Enter the PIN once.

This is the main control panel and it is built to be driven by keyboard. The buttons all say what they do; this chapter covers the order you use them in.

## The rhythm of a match

The field drives most of this by itself. What follows is what you do when the field connection is working, which is the normal case.

| WHEN                      | THE DESK DOES THIS BY ITSELF                  | YOU DO THIS                              |
|---------------------------|-----------------------------------------------|------------------------------------------|
| Match is queued           | Shows the alliance overview with robot photos | Cut to <code>CG_MATCH</code>             |
| Field arms the match      | Score bar comes in                            | Nothing                                  |
| Match starts              | Clock runs, scores update live                | Watch the show, not the desk             |
| Buzzer                    | Clock stops                                   | Cut to a wide, or to the desk            |
| Referees finish           | Nothing                                       | If a card was given, chapter 7           |
| Score posts               | Final score screen appears                    | Cut to <code>CG_SCORE</code>             |
| Gap before the next match | Nothing                                       | Sponsor, trivia, arcade, or an interview |

**If the field connection is down**, you press those beats yourself: the match lifecycle buttons across the top of the console are Preview, Arm, Start, End, Post score, Abort. The graphics behave identically.

## Keyboard shortcuts

The console is built to be driven by keyboard, because reaching for a mouse during a match is how you miss the thing you were watching for.

The match beats, along the number row:

| KEY | DOES                       |
|-----|----------------------------|
| 1   | Preview: alliance overview |
| 2   | Arm: score bar in          |
| 3   | Start the match clock      |
| 4   | End: buzzer                |
| 5   | Post the score             |
| 0   | Abort                      |

Everything else worth learning on day one:

| KEY   | DOES                                                |
|-------|-----------------------------------------------------|
| Space | <b>Mark a replay moment.</b> The one to learn first |
| T     | Show the lower third (the name strip)               |
| Y     | Hide it                                             |
| B     | Blank the program screen                            |

The next keys are only for scoring by hand, when the field connection is down. Red on the left of the keyboard, blue below it, the way they sit on the field:

| KEY | RED       | KEY | BLUE      |
|-----|-----------|-----|-----------|
| Q   | +1 fuel   | A   | +1 fuel   |
| W   | +5 fuel   | S   | +5 fuel   |
| E   | +10 tower | D   | +10 tower |

Anything you type this way renders **outlined** on air, so the audience is never shown a guess dressed as a result. See chapter 8.

Camera scenes are buttons on the console rather than keys, because cutting to the wrong camera by fumbling a key is worse than reaching for a mouse.

The console never steals a key while you are typing in a box, and it leaves Ctrl and Alt combinations to the browser: Ctrl+O resets zoom rather than aborting the live match.

## Putting people on camera

When somebody sits down for analysis or an interview:

1. In the **On camera** section, tick their name in **Saved profiles**. If they are new, type their name and role and press **Add**.
2. Check the order in **On air, in screen order**. Seat 1 is the audience's **left**. Use the arrows to move people.
3. Press **Put on air**.

Their name cards appear across the bottom of the screen, spaced evenly.

**If any of them is a student, tick "Student"**. Their card then shows a first name and a family initial ("ALEX R.") instead of a full name. Everyone in the hall still knows exactly who they are; a search engine does not, and the video is public forever. Adults doing a job under their own name (mentors, coaches, the commentators) stay in full.

If you put students on with no adult on the panel, the console tells you so. Check the shot before you cut to it: YouTube can disable chat or take a stream down over minors on camera without a visible adult, and getting that wrong ends the stream, not the clip.

Press **Clear** when they get up.

## Status cards and safety messages

Two different things, deliberately.

- **Status card**: "Field delay", "Back in ten minutes". Explains a pause. Retires itself.
- **Safety message**: covers every screen in the building, and stays until you clear it. This is for an evacuation or a hold. It is not for a long queue.

Both go to the program, the side screens and every phone at once. When you send a safety message the console goes red so that whoever walks up next can see one is live.

## Awards

Awards involve a second code, and it is not yours.

The **Judge Advisor holds the awards code**. All day, as judging wraps up, the JA loads each award's winner from their own page ( `ADDRESS/s/awards` , which takes only their code, not the desk PIN). While that is happening, your Awards section shows a lock: you can see the award titles and nothing else. That is deliberate. Nobody at the desk can see, type, or reveal a winner, so nobody at the desk can leak one, even by accident, even with a camera pointed at the console.

**Right before the ceremony, the JA gives you the code**. Type it into the Awards section's unlock box. Then, for each award:

1. Pick the award from the list. Ones marked **winner loaded** are ready: the JA's answer rides along invisibly.
2. Press **Show the award**. The title and definition go up; the GA reads it.
3. On the GA's cue ("...and the winner is"), press **Reveal the winner**.
4. Press **Clear** before the next one. Presented awards tick off the list.

The winner box on the desk is only for last-second corrections (the envelope on stage disagrees with what was loaded); leave it blank otherwise. The winner genuinely does not exist anywhere the audience can reach until Reveal, so there is nothing to leak. The lock means that now includes this console.

If no awards code was set up, there is no lock and the panel simply works on the desk PIN: type the winner yourself before Show, and do not read it aloud to anyone standing behind the desk.

## Slides and shout-outs

Slides are the between-matches cards: thank-yous for volunteer crews, session announcements, event info. The side screens rotate the whole deck on their own all day; **Next slide** steps through the same deck on the program when you want one on the big screen.

To add one during the event: pick a kind, type a title and lines (separate lines with `|`), press **Add**. It is saved immediately and survives a desk restart.

The **shout-out queue** below it fills from `/s/gp` — a phone page where anyone in the stands can submit a Gracious Professionalism moment they saw. **Nothing from that queue reaches any screen until you press Approve**. Read each one; approve the good ones into the deck; reject the rest. Nobody is notified of a rejection.

## The event timer

**Field setup 2:00** starts a countdown that takes over the side screens in digits readable from the far end of the gym — turn one side-screen TV to face the field and the drive teams have their setup clock. It clears itself the moment the match starts. The label and minutes boxes run any other countdown: meeting starts, doors, end of lunch.

## Chapter 7 · Cards, replays, and the awkward moments

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### When a team gets a card

The field usually tells the desk, and a small mark appears next to that team's number: an outlined square for yellow, a filled block for red. Different *shapes*, not just different colours, so it survives a compressed stream.

For the announcement itself, use the **card call** screen. This puts a full-screen card up after the buzzer and before the score, so the announcer has something on air while they explain it.

1. Pick the team from the list: it only offers the six who just played.
2. Pick yellow or red.
3. Type the reason **in the words the announcer will say**. It goes on screen 200 pixels tall in front of the whole gym.
4. Press **Put the card up**.
5. Press **Clear** before the score reveal.

If the desk says the card is still up after you press Clear, press Clear again. It will tell you if it failed; it will not leave you guessing.

### Replays

Press `Space` any time something interesting happens. You do not need to be exact: the mark lands a couple of seconds *before* you pressed, because everybody presses late.

The replay console ( `/s/replay` ) lists the marks. The desk has also marked things on its own: scoring bursts, lead changes, the end of auto.

### Surrogates

If a team is playing a surrogate match, one that does not count for their record, the graphics say so. Nothing for you to do; it is there so the audience is not told a team lost a match that was never theirs to lose.



## Chapter 8 · The talent view (on-air talent)

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Open `/s/talent` on a tablet. It asks for the PIN once.

Everything on this page is something you can say out loud. That is the design rule: no jargon, no raw numbers that need translating.

### What is on it

- **Who is playing**, with team names, records, and ranks.
- **What each alliance needs for a ranking point**, in a sentence. Not "ENERGIZED: 87/100" but "thirteen more fuel for Energized".
- **Pronunciation notes**, where somebody has written one in.
- **Cards a team is carrying**, so you are not surprised.
- **Whether a team is a surrogate**, so you do not say the wrong thing about their record.
- **How the day is running**, ahead or behind schedule, in minutes.

### A number with an outline around it

If a score renders as an **outline instead of solid**, it means the desk is *guessing*, usually because the field connection dropped and somebody at the desk is typing scores in by hand.

**Do not read an outlined number as final.** Say "unofficial" or wait. When the real score posts it goes solid.

This is the single most important thing on this page. The graphics will never present a guess as a fact, and neither should you.

### Interviews

The desk manager puts your guests' names on screen. Give them a moment to do it before you throw to the interview: the cards animate in, and starting the question before the name lands means the audience never learns who is talking.

# Chapter 9 · The field connection (field tech)

The desk can read the scoring system so the graphics follow the real match without anyone pressing anything.

## It is read-only, by construction

The desk **cannot** start a match, abort a match, change a score, or touch the field in any way. That is not a policy: the connection is built so that those requests do not exist. You can hand this document to the FTA and the head referee and that sentence is the whole of it.

## Setting it up

Usually nothing. The launcher looks for the field on the standard event network when it starts, finds it, and says so.

If it does not:

- 1. Check the desk laptop is on the **same network** as the scoring system. Event Wi-Fi and the field network are often not the same thing.
- 2. Check the scoring computer is on and its display is up.
- 3. Restart the desk (close the window, double-click again). The scan runs at start.

## If it stays down

The show runs anyway. Tell the desk manager, who switches to driving the match beats by hand (chapter 6). Everything looks identical to the audience except that scores are typed rather than read. Those render outlined, so nobody is misled.

This is worth rehearsing once on Friday. A field connection that drops during finals is a much better experience for everyone if the desk manager has done it before.

## The other connections you own

| THING             | HOW YOU CHECK IT                           | IF IT IS DOWN                                        |
|-------------------|--------------------------------------------|------------------------------------------------------|
| <b>OBS</b>        | The doors check, and the OBS window itself | Chapter 4                                            |
| <b>Stream</b>     | YouTube Studio shows it live               | Restart the stream in OBS; the desk is unaffected    |
| <b>Recording</b>  | Doors check says Recording                 | Replays and videos stop; the live show is unaffected |
| <b>Disk space</b> | Doors check says Disk, in <i>hours</i>     | Free space now                                       |

## Chapter 10 · Trivia, the arcade, and the music

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### Crowd trivia

Open `/s/triviadesk`. The job is one button, and its label always tells you what the next press does: **Open question** puts it on the board and starts the clock, then it reads **Reveal** (with the seconds left, so you know when the window closes), then **Next**. Press it when it says the thing you want.

The crowd joins on their phones from the QR code the trivia screen shows. There are more than thirty questions loaded already, in rounds, so you can play a round after one match and pick up a different round after the next. Scores carry across the day.

A round that has been played through will refuse to replay: that is on purpose, because replaying one pays everybody twice and quietly corrupts the leaderboard.

Names the crowd types are screened before they go on screen. Some will still get through that a reasonable person would not want on a projector; you can remove any name from the host console.

### The arcade

Open `/s/arcadedesk`. Score the game sets by hand: two players head-to-head, or three-to-four free-for-all. The overlay shows the standings.

Each info box on the arcade overlay can be shown or hidden on its own, so you can clear the screen when the game itself is the thing to watch.

### House music

**This is the one part that runs on a different computer.**

The music machine is whatever laptop is plugged into the hall's PA. Open `/s/house` **on that machine** and leave it open.

Why a separate machine: music played in the hall must not reach the stream. A recorded track on a YouTube video can get the whole archive muted or blocked, and the archive is what teams watch afterwards. Keeping the music on a machine that is not wired to the stream makes that structural rather than a thing somebody has to remember at 4pm on a Sunday.

The page refuses to run inside OBS for the same reason. If you see it refuse, that is it working.

The manager does all of this from the desk console, without leaving their seat: pause the music, resume it, duck it under an announcement, fire a team's walk-up song, or hand the room to the video game audio.

# Chapter 11 · When something goes wrong

Ordered by how often it happens.

| WHAT YOU SEE                      | ALMOST CERTAINLY                                          | FIX                                                    |
|-----------------------------------|-----------------------------------------------------------|--------------------------------------------------------|
| A screen says "reconnecting"      | The desk laptop went to sleep or the window got closed    | Wake it / start it again. Screens recover on their own |
| The overlay is frozen             | The Browser Source has "shutdown when not visible" ticked | Chapter 4, step 3                                      |
| Scores are not updating           | Field connection dropped                                  | Chapter 9. Drive it by hand meanwhile                  |
| A score has an outline around it  | The desk is guessing, and saying so                       | Nothing. It goes solid when the real score posts       |
| Camera buttons do nothing         | OBS is not connected, or a scene is misspelled            | Chapter 4, steps 2 and 7                               |
| Music is coming out of the stream | The house page is on the wrong machine                    | Chapter 10: this one matters                           |
| No robot photos                   | None uploaded for those teams                             | Fine. It falls back to a gold team number              |
| A wrong name on screen            | Somebody typed it                                         | Fix it in the profile list and press Put on air again  |
| "Port already in use" on start    | An old copy is still running                              | The launcher offers to stop it. Say yes                |

## The universal fix

Close the black window. Double-click the desk again. It comes back in a few seconds and remembers the day: cards, coverage, the run of show, sponsor counts. Every screen in the building reconnects on its own.

You do not lose the day by restarting. That is deliberate, and it was built because laptops get their power cables kicked out.

## When to escalate

Get the content desk lead for: anything involving credentials, uploads going to the wrong channel, or a message on screen that should not be there and will not clear. Everything else on this page is yours to fix.

# Appendix A · Every screen, in one table

Replace ADDRESS with what the desk window printed.

## On air: these go into OBS

| SCREEN             | URL               | NOTES                                    |
|--------------------|-------------------|------------------------------------------|
| Program overlay    | ADDRESS/s/program | 1920×1080 Browser Source. The scoreboard |
| Telestrator render | ADDRESS/s/tele    | Transparent. Layer over the replay       |
| Arcade overlay     | ADDRESS/s/arcade  | Over the game capture                    |
| Trivia overlay     | ADDRESS/s/trivia  | Crowd game                               |
| Side screen        | ADDRESS/s/side    | For venue TVs, not for OBS               |

## Run the show: these need the PIN

| SCREEN                | URL                  | WHO USES IT                                      |
|-----------------------|----------------------|--------------------------------------------------|
| Desk console          | ADDRESS/s/desk       | Desk manager                                     |
| Talent view           | ADDRESS/s/talent     | On-air talent                                    |
| Replay console        | ADDRESS/s/replay     | Replay operator                                  |
| Telestrator pad       | ADDRESS/s/draw       | Analyst's tablet                                 |
| Arcade console        | ADDRESS/s/arcadedesk | Arcade scorer                                    |
| Trivia host           | ADDRESS/s/triviadesk | Trivia host                                      |
| House audio player    | ADDRESS/s/house      | On the music machine only                        |
| Judge Advisor: awards | ADDRESS/s/awards     | JA only. Takes the awards code, not the desk PIN |
| Head referee review   | ADDRESS/s/var        | Head referee. Read-only                          |
| Phone remote          | ADDRESS/s/remote     | Anyone away from the desk                        |

## Before the event

| SCREEN           | URL             | WHO USES IT                  |
|------------------|-----------------|------------------------------|
| Team media       | ADDRESS/s/media | Whoever has the robot photos |
| Post-match cards | ADDRESS/s/cards | Social media                 |

**For everyone: no PIN**

| SCREEN           | URL             | NOTES                                                           |
|------------------|-----------------|-----------------------------------------------------------------|
| Pick a screen    | ADDRESS/s/watch | Hand this to anyone setting up a monitor                        |
| When do we play? | ADDRESS/s/next  | Parents, on their own phones                                    |
| Trivia play      | ADDRESS/s/quiz  | The crowd's phones                                              |
| Shout-outs       | ADDRESS/s/gp    | Submit a Gracious Professionalism moment. Moderated at the desk |

## Appendix B · The Friday setup list

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Print this. Tick it off.

### AV rack

- ☐ Desk laptop on the event network, plugged into power
- ☐ CalGamesContentDesk.exe copied across and run once
- ☐ Address and PIN written on the whiteboard
- ☐ Field connection confirmed (chapter 9)

### OBS

- ☐ OBS installed
- ☐ Six scenes created, names checked twice (chapter 4, step 2)
- ☐ Program overlay added to CG\_MATCH , both boxes unticked
- ☐ Overlay pasted as *reference* into CG\_SCORE and CG\_REPLAY
- ☐ Stream key entered, test stream run and stopped
- ☐ Recording tested: record thirty seconds and play it back

### Screens

- ☐ Test card checked on every display: corners visible, circle round, clock ticking
- ☐ Side screens up, full screen, rotating
- ☐ Pit monitors up
- ☐ Trivia monitor up, QR readable from where the crowd stands

### Audio

- ☐ Music machine identified and /s/house open on it
- ☐ Announcer mics tested through the stream, listening on a phone
- ☐ Confirmed music does **not** appear on the stream

### People

- ☐ Everyone has read their chapter
- ☐ Desk manager has run a practice match end to end
- ☐ Desk manager has run one match with the field disconnected, on purpose
- ☐ Talent has seen the talent view and knows what an outlined number means

### Content

- ☐ Robot photos uploaded ( /s/media )
- ☐ Sponsors entered

- ☐ Run of show entered
- ☐ Walk-up songs in place for playoff introductions
- ☐ Award titles and definitions entered, in ceremony order
- ☐ Awards code set and given to the Judge Advisor (and to nobody else); JA shown their page ( </s/awards> )
- ☐ Recognition and info slides entered (setup crew names, session times)



## Appendix C · Words this guide uses

| WORD                  | MEANS                                                                        |
|-----------------------|------------------------------------------------------------------------------|
| <b>Browser Source</b> | An OBS source that shows a web page. How the graphics get on the stream      |
| <b>Scene</b>          | One arrangement of pictures in OBS. You switch between them                  |
| <b>Overlay</b>        | The graphics on top of the camera picture: score bar, clock, names           |
| <b>Lower third</b>    | The name strip along the bottom                                              |
| <b>Program</b>        | What the audience is seeing right now                                        |
| <b>Cut</b>            | Switch instantly from one scene to another                                   |
| <b>On deck</b>        | The match after the one being played                                         |
| <b>Surrogate</b>      | A team playing an extra match that does not count for their record           |
| <b>Luma key</b>       | A way hardware switchers lay graphics over video, using black as transparent |
| <b>Doors</b>          | When the audience is let in. The deadline the pre-show check exists for      |

*This guide is part of the CalGames Content Desk, which is open source under the Apache 2.0 licence.  
Corrections are welcome and genuinely useful: if something here was wrong or confusing while you were  
standing in a gym trying to make it work, that is a bug in this document.*